Satisfiable Ramsey Diagonals Beyond Every Threshold
A Counterexample to Conjecture 00000003996

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Abstract

The proposed diagonal $n = k^2$ does not supply contradictory Ramsey CNFs, even eventually. For every threshold there is a larger k and a real two-coloring of the complete graph on k^2 vertices with no monochromatic k-clique. The standard Ramsey CNF is therefore satisfiable and admits no resolution refutation, so the claimed matching finite upper bound cannot hold. Our unbounded family uses lexicographic powers of the five-cycle and an actual finite injection. The pinned Lean project proves both color clique bounds, the cardinality and restriction bridges, satisfaction of every standard CNF clause, general resolution soundness and a closed negation of eventual refutability. A single small counterexample is not used as a substitute for the asymptotic argument.

1 The exact claim and the clause being refuted

Both original languages describe a formalized contradictory Ramsey principle, a resolution length lower bound $2^{\Omega(k)}$, and the selection $n = k^2$ making upper and lower exponent constants match. We use the ordinary two-edge-color Ramsey counterexample CNF: it forbids every all-red and every all-blue k-vertex subset of the complete graph on n vertices. A satisfying assignment is precisely a coloring with no monochromatic k-clique.

Any finite asymptotic resolution upper bound for this selected diagonal necessarily entails

$$\exists K\, \forall k \geq K: \quad \mathrm{RAM}(k^2, k) \text{ has a finite resolution refutation.} \tag{1}$$

We prove the negation of this necessary clause. We do *not* claim the lower bound alone is false: a satisfiable formula has no refutation, so a lower-bound-only convention assigning infinite minimal length would not produce a contradiction. It is the asserted contradictory status and finite matching upper bound that fail. Conditioning the diagonal on $n \geq R(k, k)$, or selecting only an unspecified unsatisfiable subsequence, would be a different statement. The unchanged exact bilingual conjecture and its Git/SHA-256 identities are included.

2 Actual recursive graphs and both clique bounds

Let C_5 be the cycle on digits $\{0, 1, 2, 3, 4\}$, with adjacent consecutive digits modulo five. Both C_5 and its complement have clique size at most two; the formal project checks this for all actual finite digit subsets, not by assigning a clique number.

Define W_0 to contain the empty word, and $W_{t+1} = \{0, 1, 2, 3, 4\} \times W_t$. Then $|W_t| = 5^t$. The red graph G_t is the following real simple graph on W_t: for different words, inspect the first digit where they differ; the edge is red exactly when those two digits are adjacent in C_5. Equivalently, at unequal first digits use C_5 adjacency, and inside an equal-first-digit block use the suffix graph G_{t-1}. The blue graph is the actual complement, not an unrelated graph with an assigned bound.

We prove by induction that every red or blue clique has at most 2^t vertices. At $t = 0$ the only vertex makes the bound 1 immediate. At the successor step, the first digits occupied by a monochromatic clique form a clique in C_5 or its complement, hence occupy at most two blocks. Within each block, suffixes form a clique of the same color and have size at most 2^t by induction. The image/fiber cardinality sum therefore gives $2 \cdot 2^t = 2^{t+1}$. Lean proves this generic block lemma and the induction for *every actual clique*, rather than enumerating finitely many word lengths.

3 The genuine diagonal restriction and unbounded parameters

For $t \geq 2$ put $k_t = 2^t + 1$ and $n_t = k_t^2$. There are enough words:

$$(2^t + 1)^2 \leq 5^t.$$

The base case is $25 = 25$. In the induction step, writing $a = 2^t$, the exact nonnegative difference is

$$5(a+1)^2 - (2a+1)^2 = a^2 + 6a + 4 \geq 0.$$

Thus the successor inequality follows from the preceding one and $5^{t+1} = 5 \cdot 5^t$.

Choose an actual injection from $\mathrm{Fin}(n_t)$ into W_t and pull back G_t. The formal choice uses the proved finite cardinality comparison and standard `Classical.choice`; no prescribed lexicographic initial segment is assumed or claimed as a computed output. Clique cardinalities cannot increase under an injective pullback, and complement commutes with that pullback. Hence this genuine graph on exactly k_t^2 vertices has no red or blue k_t-clique, since $2^t < k_t$.

The parameters k_t are unbounded. In particular, for every natural threshold K the project proves existence of $k \geq K$ and an actual graph on $\mathrm{Fin}(k^2)$ whose red graph and actual complement are both k-clique-free. This covers the eventual/asymptotic quantifier, not merely a finite-prefix failure.

4 Standard CNF and ordinary resolution are formally connected

The edge variables are the actual unordered edges of the complete graph, excluding loops. For every k-subset S, the CNF contains the two clauses

$$\bigvee_{e \in \binom{S}{2}} \neg x_e, \qquad \bigvee_{e \in \binom{S}{2}} x_e.$$

They forbid respectively the all-red and all-blue coloring. For a real graph g, assign x_e true exactly when its edge lies in g. If g and g^c are both k-clique-free, every k-subset has an edge of each color. The project proves that this actual assignment satisfies *every generated clause*, with no omitted subsets, arbitrary named variables or unused forbidden patterns. Repeated clauses may be deduplicated without changing truth.

Clauses are finite signed-literal sets. Ordinary resolution removes opposite pivot literals from the two earlier parent clauses and unions the remaining literals; weakening enlarges an earlier clause. The proof relation has only input, resolution and weakening constructors. For every assignment, satisfaction of the parents entails satisfaction of the resolvent. Structural induction proves soundness of any such derivation over any finite vertex set. A satisfying assignment therefore excludes derivation of the empty clause, with no bound on proof length. Finite DAG or listed proofs unroll to these trees without changing the existence of a finite proof; this standard mathematical correspondence is explained in prose, not advertised as an additional general Lean DAG theorem.

Applying soundness to the graphs from the previous section proves

$$\forall K\ \exists k \geq K: \quad \mathrm{RAM}(k^2, k) \text{ is satisfiable and has no resolution refutation.}$$

Theorem 1. *Conjecture 00000003996's finite matching-upper-bound claim for the selected diagonal $n = k^2$ is false under the standard Ramsey CNF and ordinary resolution conventions.*

Proof. An eventual finite upper bound implies (1). Given its threshold K, the actual coloring family supplies $k \geq K$ with a satisfying assignment. Soundness excludes the alleged resolution refutation. This contradicts (1). □

5 Formal scope, reproduction and limitations

`Family.red_blue_clique_card` proves the full recursive bound; `Family.diagonal_family` and `Family.beyond_every_threshold` give actual restricted graphs on the selected diagonal. `CNF.cliqueFree_implies_satisfiable` connects them to the actual CNF.

`CNF.satisfiable_excludes_refutation` applies general soundness. `necessary_finite_upper_bound_false` proves the closed necessary-clause negation; `actual_diagonal_counterexamples` exhibits the actual graph, both clique-free properties, assignment and exclusion of any refutation beyond every threshold.

The portable project pins Lean 4.33.1 and Mathlib commit

`0df444a360eaa60ab8c11dca51a86af692955474`.

Every transitive Git revision is fixed. The reproducer checks the exact bilingual source, toolchain and every dependency's actual HEAD/origin/clean tracked source, clean-builds only this project's code with warnings as errors and prints every public theorem's transitive axioms. Only `propext`, `Classical.choice` and `Quot.sound` are allowed. Exact finite regressions are supplemental, not substitutes for the unrestricted induction and threshold quantifiers. The actual PDF is compiled from this source using Tectonic 0.15.0 and `SOURCE_DATE_EPOCH=0`.

No optimal Ramsey number, general resolution-length lower bound, new proof system, finite proof-complexity value for a satisfiable CNF or merely finite $k = 4$ asymptotic refutation is claimed. This is a disproof of the source's required finite-upper-bound premise, not a disproof of Ramsey's theorem. AI-assisted implementation and a different independent Agent review are correlated AI checks, not human expert endorsement. No affiliation, first-solve credit, award or organizer acceptance is claimed.